

GCN-Enhanced Spatial-Spectral Dual-Encoder Network for Simultaneous Segmentation of Retinal Layers and Fluid in OCT Images

Guogang Cao^{a,*}, Zhilin Zhou^a, Yan Wu^a, Zeyu Peng^a, Rugang Yan^a, Yunqing Zhang^a, Bin Jiang^b

^a School of Computer Science & Information Engineering, Shanghai Institute of Technology, Shanghai 201418, China

^b Department of Ophthalmology, Zhongnan Hospital of Wuhan University, Wuhan 430061, China

ARTICLE INFO

Keywords:

Medical image segmentation
U-Net
Fast Fourier Transform
Graph convolution
OCT Retinal images
Macular Edema

ABSTRACT

Changes in retinal thickness and the morphology of fluid provide valuable diagnostic information for Macular Edema (ME). Optical Coherence Tomography (OCT) is a pivotal diagnostic tool for ME, and the accurate segmentation of OCT retinal images aids ophthalmologists in monitoring disease progression. However, the unpredictable nature of ME's occurrence poses significant challenges for existing methods in achieving precise segmentation of both retinal layers and fluids simultaneously. In this study, a novel U-shaped encoder-decoder architecture named GD-Net, is introduced for simultaneously segmenting retinal layers of diverse thicknesses and fluids. Specifically, a novel Fast Fourier Encoder is integrated within the encoder branch to capture spectral domain features overlooked by the spatial encoder. Additionally, Multi-scale Graph Convolution module is inserted between the encoder and decoder branches to leverage retinal layer topology for spatial reasoning. GD-Net is further optimized using a joint loss function, which includes a weighted version of region segmentation loss and contour loss, effectively mitigating the issues of blurred fluid edges and internal holes. The method was validated on three publicly available datasets: DUKE DME, Peripapillary OCT, and RETOUCH, achieving overall mean Dice score of 0.839, 0.826, and 0.872, respectively, outperforming the other state-of-the-art techniques. GD-Net demonstrated consistent high-precision segmentation on B-scans from different devices, proving its robustness and surpassing other methods in terms of visual quality, thus offering substantial support for clinical ophthalmologists in diagnosing ME. The code and weights are publicly available at: <https://github.com/DBook111/GD-Net>.

1. Introduction

The Pathological fluid and retinal thickness changes are both considered biomarkers for tracking the progression of Macular Edema (ME). ME is characterized by the accumulation of fluid and inflammation in the retinal layers within the macular region. Numerous retinal diseases, such as Diabetic Macular Edema (DME), Age-Related Macular Degeneration, and Retinal Vein Occlusion, all of which may manifest ME [1–3]. The abnormal edema results in the disruption of normal retinal structures and can lead to visual blurring or blindness. Accurately segmenting the location of ME assists ophthalmologists in examination and analysis, facilitating the diagnosis and formulation of treatment plans. Besides, variations in the thickness of the retinal layers can also provide critical information for the diagnosis of diseases such as Macular Degeneration, Diabetic Retinopathy (DR), and Glaucoma [4]. For

instance, morphological changes in the Retinal Nerve Fiber Layer and Ganglion Cell Layer are instrumental in the precise diagnosis of glaucoma [5]. In the progression of DR, the central retinal thickness tends to increase concomitant with the advancement of the disease [6,7]. The choroidal thickness is closely correlated with the severity and type of ME, serving as a common indicator for early diagnosis of DME [8].

Optical coherence tomography (OCT) is an advanced imaging technique that provides high-resolution, cross-sectional images of the retinal structures. It provides images with micrometer-level resolution, allowing ophthalmologists to clearly visualize each layer of the retina and areas of edema, including subtle structural and pathological changes [9]. As demonstrated in Fig. 1(a), OCT can identify three types of pathological fluids. Intraretinal Fluid (IRF) typically presents as fluid accumulation within the retinal layers, leading to retinal swelling or increased thickness [10]. Subretinal Fluid (SRF) is the accumulation of

* Corresponding author.

E-mail address: guogangcao@163.com (G. Cao).

<https://doi.org/10.1016/j.bspc.2024.106702>

Received 15 April 2024; Received in revised form 27 June 2024; Accepted 31 July 2024

1746-8094/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

fluid beneath the outer plexiform layer but above the retinal pigment epithelium layer. Pigment Epithelial Detachment (PED) usually appears as a protrusion between the retinal pigment epithelium layer and Bruch's membrane (BM), resulting in irregularities in this layer.

OCT images are examined and analyzed by ophthalmologists for diagnosis and treatment. With the increasing number of patients and the relative shortage of ophthalmologists, the traditional manual screening process has become both time-consuming and inefficient. Recently, deep learning technology has made significant breakthroughs in the field of retinal OCT image segmentation. Most studies focus solely on segmenting retinal layers of varying thicknesses [11,12]. Others focus on segmenting areas of pathological fluid [13–15]. Although some studies simultaneously segment both retinal layers and abnormal edema, these methods generally are incapable of distinguishing different types of pathological fluids [16,17]. Edema areas are typically characterized by fuzzy boundaries, uncertain sizes and locations, and their internal shape's prior knowledge should be a continuous area without holes. Therefore, accurately segmenting both the retinal layers and abnormal edema areas remains a research topic worthy of in-depth exploration.

We argue that images captured by OCT devices contain a wealth of spectral domain information, which plain convolutions have not fully utilized. Specifically, areas of pathological fluid, due to their distinct light reflection and absorption properties, generally manifests as low signal areas (dark regions), while PED, attributed to their unique structural and compositional characteristics, often appears as high signal areas (bright regions). The high and low signal regions indirectly reflect the differing spectral properties of tissues. Additionally, the spatial encoder, due to its limited receptive field size, can only capture local texture details. Inspired by the approach of Chi et al., which involved the replacement of vanilla convolutions with fast Fourier

convolutions [18], we introduce a novel Fast Fourier Encoder (FFE) with non-local receptive fields, tailored for learning the characteristics of fluid. Therefore, we combined the FFE with the spatial encoder to leverage the advantages of both and mitigate their respective shortcomings, thereby improving the segmentation performance. Jamal et al. elucidated the efficacy of using graphs to represent domain knowledge and the structural relationships of organizations in medical image segmentation [19]. We extended the theory of Jamal et al. to the vertically ordered retinal layers, where the spatial relationships between these layers can be abstracted and represented as graphs, as shown in Fig. 1(c). Therefore, we design a Multi-scale Graph Convolution (MGC) module for segmenting retinal layers of varying thicknesses. Recognizing the challenges posed by blurred boundaries and internal holes in the segmentation of pathological fluid, we design a novel joint loss function to mitigate these issues. To demonstrate the effectiveness of the framework, experiments are conducted on three publicly available datasets: DUKE DME [20], RETOUCH [21], and the Peripapillary OCT datasets [22]. Compared to state-of-the-art techniques and baselines, the proposed model exhibited optimal performance across all three datasets. In the future, we plan to integrate this model into the workflow for diagnosing macular edema. In this work, our contributions can be summarized as follows:

- (1) We propose a novel method, GD-Net, for simultaneous segmentation of retinal layers and fluid. Experiments conducted on three publicly available datasets demonstrate that the proposed model exhibits superior performance compared to the state-of-the-art techniques.

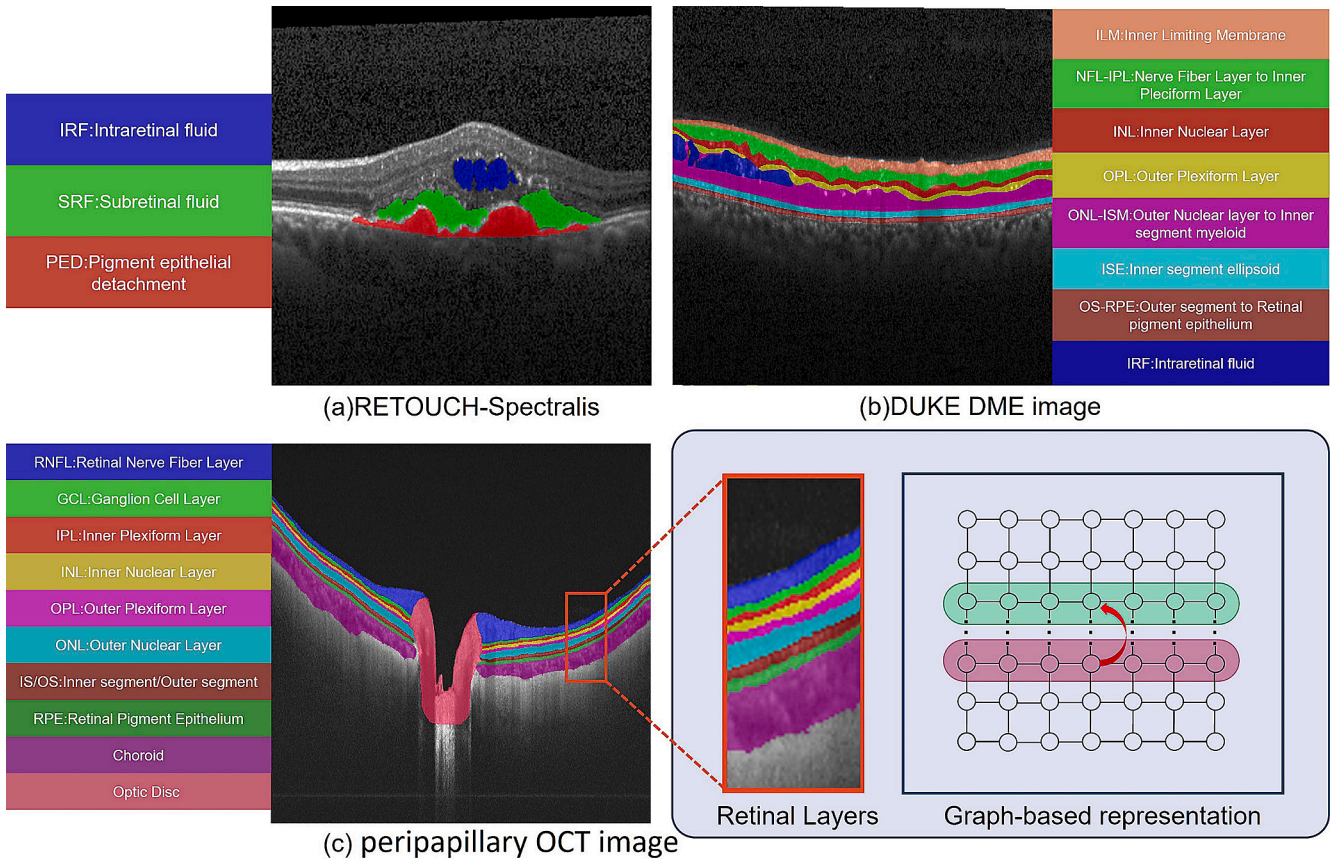


Fig. 1. Examples of annotated OCT images from the RETOUCH(a), DUKE DME datasets(b) and Peripapillary Retinal OCT datasets(c). The different retinal layers and pathological fluids are annotated by different color. The retinal layers exhibit an orderly arrangement in the vertical direction, which can be abstracted into a graph-based representation.

- (2) A novel FFE was proposed to fully capture the characteristics of fluid in pathological OCT images by including spectral features overlooked by the spatial encoder.
- (3) The MGC module was designed to leverage anatomical prior knowledge of retinal layers and employ graph reasoning to enhance segmentation performance across retinas of varying thickness.
- (4) A joint loss function scheme was developed based on region and boundary information, mitigating issues of blurry boundaries and internal holes in fluid segmentation.

2. Related work

2.1. Retinal segmentation methods

Methods for retinal segmentation can be categorized into traditional approaches and deep learning-based ones.

Traditional segmentation methods rely on mathematical models. The retinal layering structure is detected by analyzing anatomical features and incorporating clinical prior knowledge. OCT images are commonly construed as constituents of Graph Construction (GC), with problem-solving facilitated through the application of Dynamic Programming (DP) techniques [20,23]. Noteworthy is the work of Chiu et al. [24], who leveraged graph theory in tandem with DP to address the challenge of identifying shortest paths within OCT retinal images, thus deriving estimates of layer boundaries. It was further refined in subsequent methods, incorporating soft constraints within the algorithm to introduce prior information learned from the model in the GC [25]. Further strides were taken by Tian et al., as they devised graph-theory-based boundary detectors tailored for the rapid and precise segmentation of choroidal boundaries [26]. Srinivasan et al. proposed the utilization of support vector machines, graph theory, and DP within GC for the automated segmentation of retinal layers in spectral domain OCT images [27].

In recent years, the advancement of Convolutional Neural Networks (CNN) diverged from traditional methods. CNN found extensive application in segmenting images acquired from various sources, yielding numerous exciting achievements. Some recent approaches tended to employ neural networks for end-to-end segmentation. The Fully Convolutional Network (FCN) [28] and U-Net [29] stood out as the earliest and most renowned autoencoder-based frameworks for two-dimensional medical image segmentation. Roy et al. introduced ReLayNet, marking the pioneering application of deep learning to automatic segmentation of retinal OCT images [16]. Cao et al. employed A-Scans data for single-step regression training to obtain the boundary of retinal layer. They achieved state-of-the-art performance while utilizing less labeled dataset [30]. Building on this, Cao et al. further employed the transformer's self-attention mechanism within a U-shaped network to convert two-dimensional features into one-dimensional, thereby enhancing the model's ability to capture long-range contextual information [31]. However, this approach was specifically designed for certain retinal layers, and its generalizability to pathological fluids was not validated. Rasti et al. introduced RetiFluidNet, which was capable of automatically extracting deformation-aware representations at various levels. However, this method was specifically designed for fluid segmentation and did not exhibit satisfactory segmentation performance at the retinal layer [13].

2.2. Spectral neural networks

The spectral domain was previously used solely for accelerating convolutions, but it now also provides robust support for constructing deep networks. For example, Chi et al. [32] introduced spectral residual learning, employing Fast Fourier Convolution to map features into the spectral domain for learning. This method addressed the historical issue of insufficient receptive field coverage in a more efficient manner.

Rippel et al. trained a powerful convolutional neural network (CNN) in spectral domains. They demonstrated the effectiveness of complex-coefficient spectral parameterization of convolutional filters [33]. By using tools from Fourier analysis, Rahaman et al. highlighted a learning bias of deep networks towards low frequency functions [34]. Recently, Suvorov et al.'s application of Fast Fourier Convolution [18] in image restoration tasks has sparked extensive discussions [35]. They exploited global patterns present in OCT images that vanilla convolutions struggle to extract effectively. Building upon the previous work, Farshad et al. proposed the Y-Net, introduced a Fast Fourier Encoder to enhance the model's sensitivity to fluid regions within images, thus improving fluid segmentation accuracy [17].

Different from prior works, we devised a novel Spectral Transcoder within the framework of Fast Fourier Convolution, which learns spectral features related to the fluid accumulation position in OCT retinal images. Comprehensive empirical results provide validation for the efficacy of our proposed module.

2.3. Graph convolution networks

Graph-structure models have recently been extensively employed in segmentation tasks for its ability to capture the inherent graph structure and relationships between connected nodes. Meng et al. designed a graph neural network that leverages both region and boundary features, enabling simultaneous reasoning and aggregation of multiple features at different levels, which demonstrates a significant improvement in optic disc and optic cup segmentation [36]. Shin et al. employed GNN to learn the global structure of container shapes, which reflects the connectivity of adjacent vertices [37].

Similarly, Chen et al. introduced global graph reasoning units that efficiently infer global relationships between distant regions within an image [37].

The above-mentioned methods utilize GNN to address the issue of long-distance intra-domain feature propagation. It directly performs graph convolution over the entire region, where different nodes represent regions corresponding to different pixels. Li et al. devised a multi-scale, two-stage segmentation network that rigorously adheres to anatomical priors of retinal layers [22]. However, their direct utilization of max-pooling operations to reduce the dimensions of feature maps results in information loss. In contrast, owing to the varying thickness of each retinal layer, we employ pooling kernels of different sizes to create multiple scales and utilize graph convolutional network (GCN) to learn their inter-domain relationships.

3. Method

The framework of the proposed segmentation method is illustrated in Fig. 2. Initially, preprocessing operations including edge enhancement and histogram matching are applied to the input images. Subsequently, the GD-Net is employed to segment the preprocessed images. A contour loss function is introduced to mitigate the issue of data imbalance, as the size of the pathological fluid regions is several orders of magnitude smaller than that of the retinal layer areas. Furthermore, the contour loss can eliminate unnecessary holes inside the pathological fluid, aligning with prior information [14,38]. For experiments related to retinal segmentation task, as demonstrated in Appendix B.

3.1. Preprocessing of OCT B-scans

For three different datasets, we utilized the same preprocessing methods: edge enhancement [39] and histogram matching. To enable contour loss to more effectively penalize boundary features, we introduced edge enhancement to enhance feature discrimination, thereby reducing the model's dependency on a large volume of training data. Histogram matching involves adjusting the histogram of one image to match the histogram of another reference image. This technique ensures

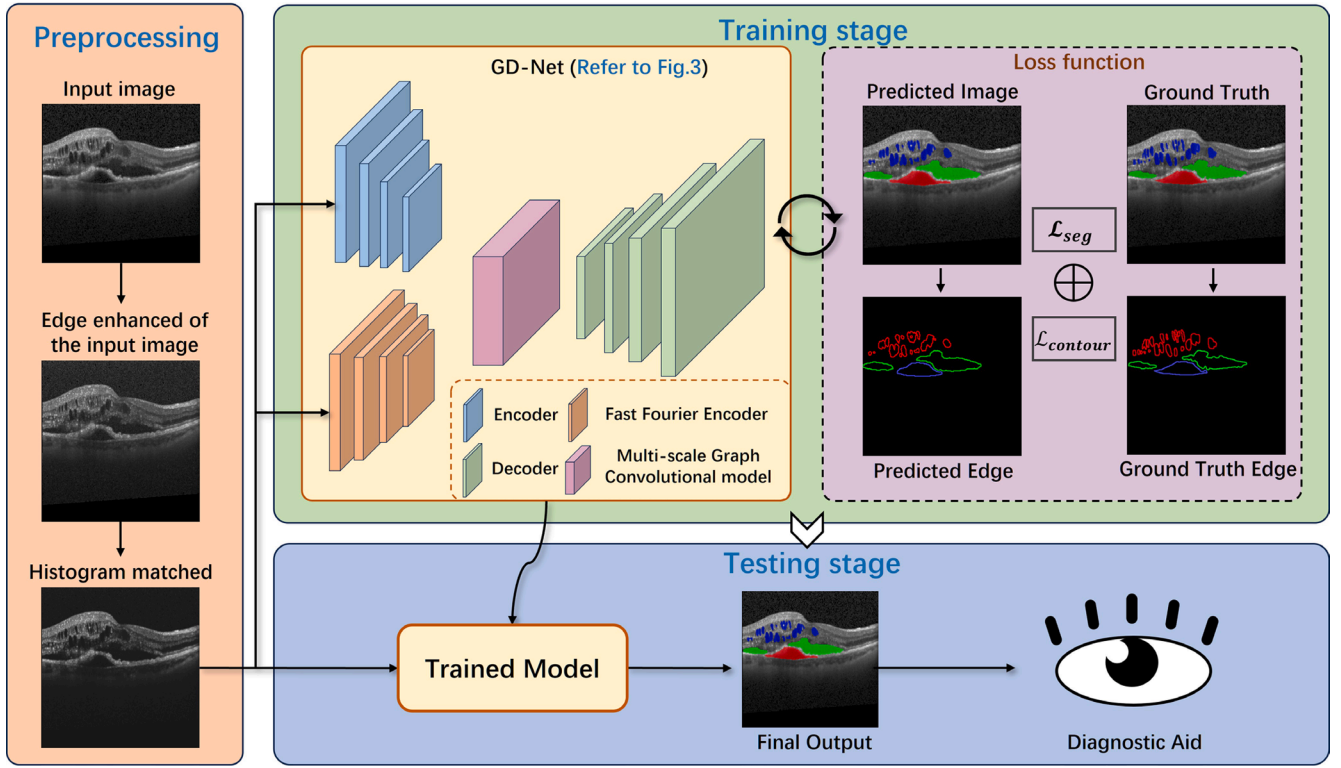


Fig. 2. Flowchart of the Proposed Method for Segmenting the Retinal Layer and Pathological Fluid.

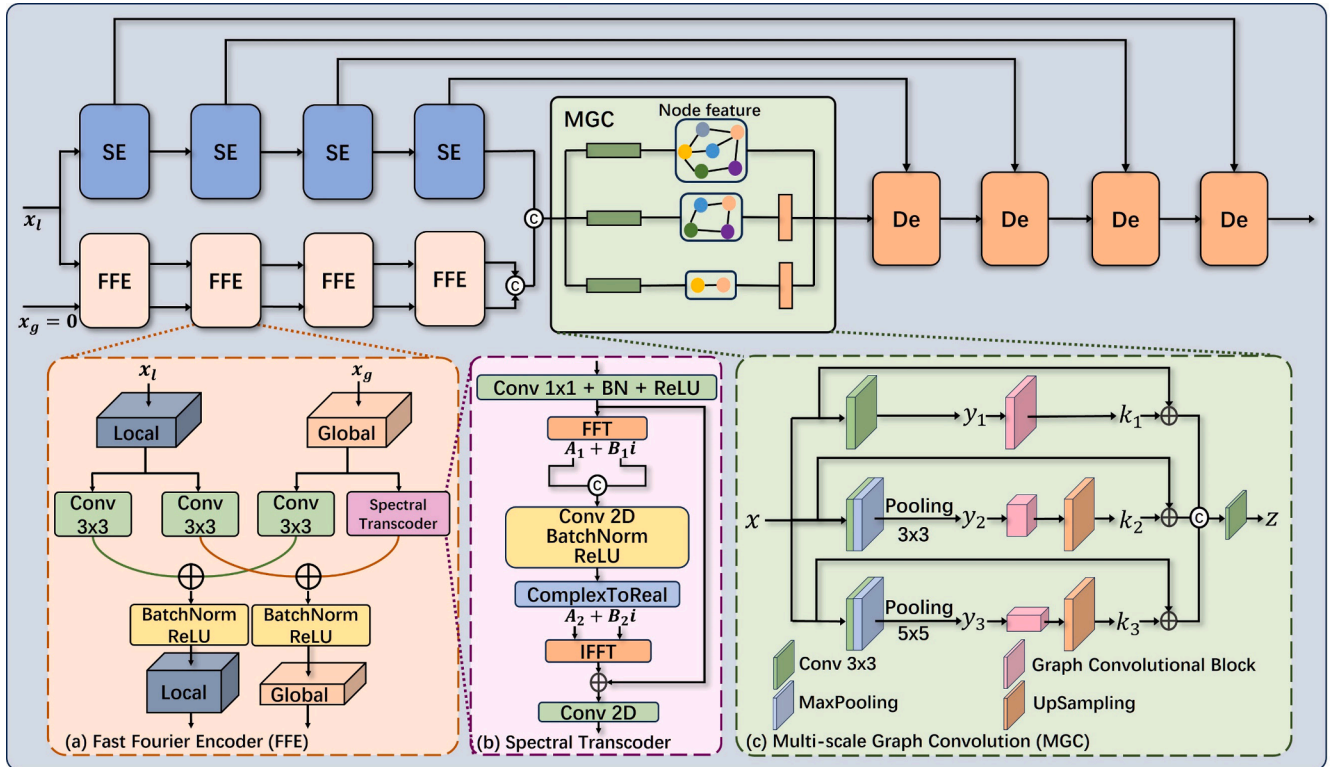


Fig. 3. The architecture of the proposed GD-Net. Here, x_l as the input for local features, x_g serves as the input for global features. SE denotes the Spatial Encoder, and De represents the Decoder. (a) FFE: The FFE employs Conv2D to process local features, and utilizes a Spectral Transcoder to handle global features. (b) Spectral Transcoder: The input features undergo a Fast Fourier Transform (FFT) and Conv2D processing to obtain frequency domain features. Finally, the processed features are transformed back to the spatial domain using the Inverse FFT. (c) MGC: The structure of the MGC consists of three branches. No pooling operation is in the first branch. The second and third branch incorporate pooling operations with 3×3 and 5×5 kernels, respectively.

that the brightness and contrast of different images are consistent, which is crucial for reducing variability caused by illumination differences. By applying histogram matching, the contrast of the images in the dataset was enhanced. Enhanced contrast makes it easier for the model to identify and focus on critical features, thereby improving its performance. This preprocessing step ensures that the model is not distracted by variations in lighting and instead concentrates on the structural and textural aspects of the images.

In our implementation, we selected one image as reference histograms and matched the histograms of other images in the dataset to these references. This approach provided uniformity across the dataset, making regions of interest (ROI) more noticeable, and thereby increasing the accuracy of segmentation.

3.2. Network architecture

As illustrated in Fig. 3, GD-Net is an improved U-Net network architecture. It consists of two encoder branches, a novel MGC and a decoder branch. The architectures of the Spatial Encoder and Decoder are shown in Fig. 4. The spatial encoder branch is combined with the spectral encoder branch to enhance feature extraction. MGC is inserted at the bottleneck, followed by the decoder branch.

3.3. Fast Fourier encoder

Based on the spectral convolution theorem of Fourier theory, which posits that modifying a single value in the spectral domain can globally affect the entire original data [40], and inspired by the work of Chi et al. in replacing vanilla convolution with fast Fourier convolution [18], we introduce a Fast Fourier Encoder (FFE) with global receptive field.

The initial FFE receives the same input x as the spatial encoder. The input image x serves as the x_l for the first FFE module. As the shallow encoder blocks primarily capture low-level local patterns, we configure $x_g = 0$ for the first FFE, as shown in Fig. 3. The subsequent three inputs of FFE are the outputs of the preceding encoder. Specifically, the $x_l \in \mathbb{R}^{1 \times H \times W}$ of the first FFE block is the feature map of the input image, x_g is 0, and its outputs are $x_l' \in \mathbb{R}^{3C \times H \times W}$ and $x_g' \in \mathbb{R}^{C \times H \times W}$. The inputs to the

second FFE are x_l' and x_g' , and the outputs are $x_l'' \in \mathbb{R}^{6C \times \frac{H}{2} \times \frac{W}{2}}$ and $x_g'' \in \mathbb{R}^{2C \times \frac{H}{2} \times \frac{W}{2}}$. The channel ratio of global to local features is 1 : 3. The last three FFE blocks halve the feature map size and double the number of channels. After passing through four FFEs, the outputs of x_g and x_l are concatenated along the channel dimension.

As shown in Fig. 3(a), the FFE divides features into local and global branches. One of them is the Spectral Transcoder path denoted as $f(x)$, while the other three are vanilla convolution paths denoted as $g(x)$. The vanilla convolution has a small receptive field due to the 3×3 convolution kernel, while the global path possesses a global receptive field due to the presence of the Spectral Transcoder. Vanilla convolution and the Spectral Transcoder extract information at different scales. Subsequently, in order to fully utilize multi-scale receptive fields, these four paths undergo multi-scale fusion within the FFE. The fusion process can be described using the following equations:

$$Y_g = f_{g \rightarrow g}(x_g) + g_{l \rightarrow g}(x_l) \quad (1)$$

$$Y_l = g_{g \rightarrow l}(x_g) + g_{l \rightarrow l}(x_l) \quad (2)$$

where $g_{g \rightarrow l}$ and $g_{l \rightarrow g}$ paths serve as fusion paths, while the $g_{l \rightarrow l}$ path employs a vanilla convolutional encoder. The highest computational complexity lies within the $f_{g \rightarrow g}$ path, which we refer to as the Spectral Transcoder.

As depicted in Fig. 3(b), upon the application of the Fast Fourier Transform (FFT) to real-valued tensors, conjugate-symmetric Hermitian matrices A_1 and B_1 emerge. To alleviate computational burden, the complex-valued tensor is decomposed into its real and imaginary components, subsequently concatenated along the final dimension. After extracting features in the spectral domain using 1×1 convolution, the process is followed by Batch Normalization and ReLU activation. Subsequently, through a complex transformation, the results are restored to their complex representation. Ultimately, the features are transformed from the frequency domain to the spatial domain using the Inverse Fast Fourier Transform (IFFT).

3.4. Multi-scale graph convolution module

As illustrated in Fig. 1, the layers within the retina exhibit a striped arrangement in their topological sequence. Leveraging this knowledge prior, we devised an innovative Multi-scale Graph Convolution (MGC) module. As shown in Fig. 3(c), MGC comprising three paths with varying receptive field sizes and Graph Convolutional Block (GCB), the MGC captures multi-level vertical structures and infers interrelationships among regions. This strategy is capable of addressing the challenges presented by segmenting retinal layers of varying thicknesses. The MGC accepts features $x \in \mathbb{R}^{C \times H \times W}$ merged from spatial encoder branch and spectral encoder branch as input. It then divides these features into three distinct paths. The first path excludes pooling layers, utilizing a 3×3 convolution layer to increase feature depth, resulting in $y_1 \in \mathbb{R}^{2C \times H \times W}$, the convolution layers in the remaining two paths maintain feature dimensions and are followed by pooling layers with kernel sizes of 3×3 and

5×5 , respectively, producing $y_2 \in \mathbb{R}^{C \times \left(\left\lfloor \frac{H-3}{3} \right\rfloor + 1\right) \times \left(\left\lfloor \frac{W-3}{3} \right\rfloor + 1\right)}$ and $y_3 \in \mathbb{R}^{C \times \left(\left\lfloor \frac{H-3}{3} \right\rfloor + 1\right) \times \left(\left\lfloor \frac{W-3}{3} \right\rfloor + 1\right)}$. All three paths employ GCB. This arrangement allows the encoding of information from retina layers of varying thicknesses into a global feature map. Subsequently, the second and third paths use bilinear interpolation for upsampling to obtain $k_2 \in \mathbb{R}^{C \times H \times W}$ and $k_3 \in \mathbb{R}^{C \times H \times W}$, matching the dimensions of $k_1 \in \mathbb{R}^{2C \times H \times W}$. Each path employs residual connections, and then, these three paths are recombined, with features concatenated along the channel dimension. Finally, a 3×3 convolution layer adjusts the channels to produce the output $z \in \mathbb{R}^{C \times H \times W}$. For the specific values of y_2 and y_3 , please refer to

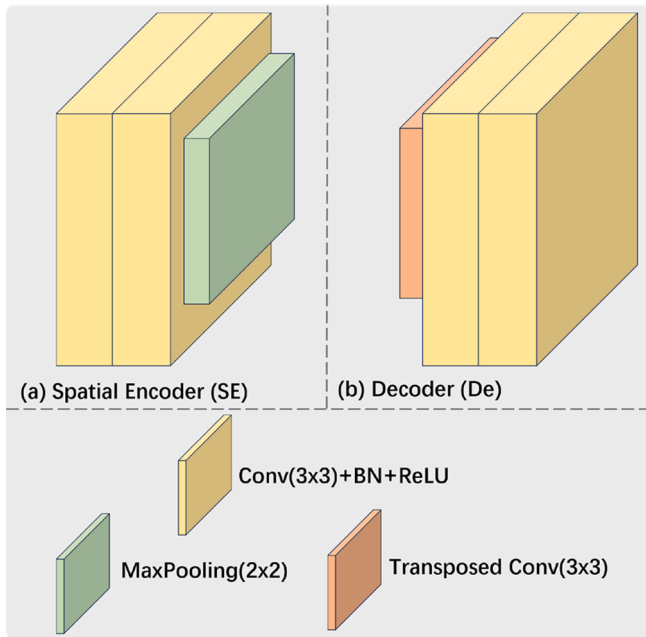


Fig. 4. (a) Spatial Encoder: SE is used to encode the spatial information of the image. (b) Decoder: De first employs a transposed convolution layer, followed by the same convolution blocks as the spatial encoder to upsample the features. The final segmentation map is generated by the last decoder block.

Appendix A. MGC is especially important because retinal layers such as the Inner Nuclear Layer (INL) are only a few pixels thick. Without employing MGC, significant boundary diffusion and unreliable layer thickness estimation may occur.

Earlier investigations have employed projection techniques to transform feature maps into region-based representations, subsequently employing graph reasoning to capture global relationships [41,42]. Inspired by graph-based global reasoning networks, GCB is crafted to adeptly extract features from both retinal and pathological fluid [22,43,44].

In this work, we employ a projection process to achieve objectives akin to those of the graph Laplacian matrix. The architecture of the GCB is illustrated in Fig. 5. Initially, the feature map $X \in \mathbb{R}^{C \times H \times W}$ is input in parallel into two convolutional blocks to generate two feature maps: one obtains the projection matrix, the other reduces dimension. Subsequently, the dimensionality-reduced feature map is reshaped into $X_a \in \mathbb{R}^{C_a \times HW}$, and the projection matrix is reshaped and transposed into $X_b \in \mathbb{R}^{HW \times C_b}$. The matrix multiplication of X_a and X_b is performed to obtain the node feature map $Q \in \mathbb{R}^{C_a \times C_b}$, which is then subjected to Graph Convolutional Network to derive the node feature map $V \in \mathbb{R}^{C_a \times C_b}$. X is further subjected to a convolution operation and reshaped to obtain $X_c \in \mathbb{R}^{C_b \times HW}$, after which X_c is multiplied by V to transform back to the original latent space. This is then reshaped and subjected to a convolution operation to acquire the final feature map $Y \in \mathbb{R}^{C \times H \times W}$. Finally, we perform an element-wise addition of X and Y to acquire the new feature map $Z \in \mathbb{R}^{C \times H \times W}$.

3.5. Loss function

Joint loss function $\mathcal{L}_{joint}$ is defined by Eq. (3),

$$\mathcal{L}_{joint} = \alpha \mathcal{L}_{seg} + \beta \mathcal{L}_{contour} \quad (3)$$

It can be observed that the joint loss function includes two terms, $\mathcal{L}_{seg}$ and $\mathcal{L}_{contour}$, with α and β serving as the weighting coefficients. The objective of the $\mathcal{L}_{joint}$ is to find appropriate weight coefficients that balance the penalties on regional localization and contour delineation.

Specifically, each of those two loss functions could be defined as follows,

$\mathcal{L}_{seg}$: A weighted combination of Dice Loss [45], Cross-Entropy Loss [46], and Focal Frequency Loss [47] is employed to train $\mathcal{L}_{seg}$. The empirical evidence demonstrates that the composite loss function yields favorable results in the segmentation of both retinal and fluid accumulation regions. $\mathcal{L}_{seg}$ is calculated with labels provided by the dataset, shown as below,

$$\mathcal{L}_{seg} = \lambda_1 \mathcal{L}_{Dice} + \lambda_2 \mathcal{L}_{CE} + \lambda_3 \mathcal{L}_{FF} \quad (4)$$

$$\mathcal{L}_{Dice} = 1 - \frac{1}{N} \sum_{i=1}^N \left(\frac{2 \sum_{x \in \Omega} p_i(x) \times g_i(x)}{\sum_{x \in \Omega} p_i(x) + \sum_{x \in \Omega} g_i(x)} \right) \quad (5)$$

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N g_i \log(p_i) \quad (6)$$

$$\mathcal{L}_{FF} = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} \omega(u, v) |p(u, v) - g(u, v)|^2 \quad (7)$$

where λ_1, λ_2 and λ_3 are the weights of each loss term. g_i and p_i respectively represent the ground truth and the probability that pixel x belongs to class i , and N denotes the number of categories to be segmented in the label. Eq. (5) represents the Dice Loss, which takes into consideration the intersection over union (IoU). Eq. (6) and Eq. (7) respectively denote the Cross-Entropy Loss and the Focal Frequency Loss. In Eq. (6), $\mathcal{L}_{CE}$ maximizes the cross-entropy information between labels and predicted results. In Eq. (7), M represents the width of the image, and N represents the height of the image. During loss computation, the loss value is divided by the total number of pixels MN to ensure that the scale of the loss function is independent of the image size. $\omega(u, v)$ is a weight in the frequency domain, used to adjust the impact of different frequency components in the loss function. It assigns different weights to each frequency component based on their position (u, v) in the frequency domain. $\mathcal{L}_{FF}$ serves to narrow the disparity between the real labels and predicted values in the spectral domain.

$\mathcal{L}_{contour}$: $\mathcal{L}_{seg}$ measures the overlap between predictions and ground truth to optimize model parameters. Sole reliance on region-based loss functions can lead to holes in pathological fluid and blurred boundaries.

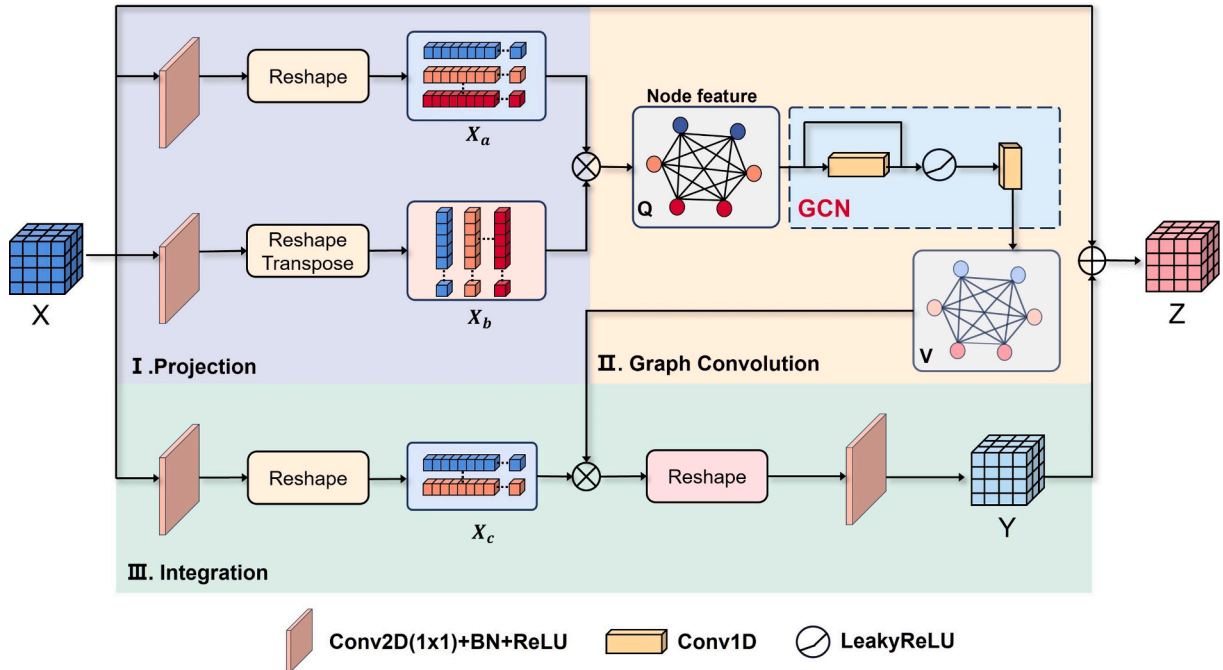


Fig. 5. Graph Convolutional Block (GCB): initially projects the input features into the node space, and subsequently conducts graph convolutions within the node space to extract comprehensive global node features. To yield the ultimate features for output, the global node features are then inverse-projected back into the original feature space.

To address this, we introduce the $\mathcal{L}_{\text{contour}}$, derived from contour pseudo-labels created via edge detection and connectivity analysis [48–50]. As illustrated in Fig. 6, the orange and white outlines are the $\text{predict}_{\text{contour}}$ and $\text{label}_{\text{contour}}$, respectively. Therefore, $\mathcal{L}_{\text{contour}}$ is defined as below,

$$\mathcal{L}_{\text{contour}} = -\frac{1}{\sigma} \sum_{i=1}^K (l_{\text{contour}}(i) \times \log p_{\text{contour}}(i)) \quad (8)$$

where σ represents the number of contour pixels, while l and p denote the ground truth and the probability associated with each contour pixel, respectively. $\mathcal{L}_{\text{contour}}$ enhances the significance of edge pixels, thereby prioritizing the optimization of model parameters towards these pixels. This enhancement increases the model's ability to distinguish edge pixels from noisy backgrounds. Moreover, it effectively addresses data imbalance, crucial in OCT images where retinal layers and pathological fluids are a small portion of the background.

4. Experiments

4.1. Datasets

We used three popular publicly available datasets in this work. Specifically, we used the DUKE DME dataset [20] and the Peripapillary OCT dataset [22] to validate the performance in segmenting retinal layers of varying thicknesses. The DUKE DME dataset and RETOUCH [21] datasets were utilized to assess the segmentation performance on pathological fluids.

The DUKE DME dataset, collected by Chiu et al., consists of images acquired using the Spectralis HRA+OCT device manufactured by Heidelberg Engineering in Germany. The dataset comprises a collection of 11 OCT B-scan images per each of the 10 patients afflicted with diabetic macular edema. We adhered to the same experimental protocol as previous studies, partitioning the dataset in accordance with a balanced 6:2:2 ratio. Additional comprehensive insights into the dataset's specifics can be referenced from [20]. It is worth noting that among the 22B-scan images from the last two subjects used for testing, not all of them contained fluid. Therefore, in the computation of the Dice Score for fluid, we could only consider those B-scan images that exhibited fluid.

The Peripapillary OCT dataset comprises a total of 61 subjects, with each subject including 12 OCT B-scans obtained through scanning by the Ophthalmology Department at Shanghai General Hospital, utilizing the DRI OCT-1 Atlantis system (Topcon Corporation, Tokyo, Japan). To ensure comparability, we adopted the same data processing approach as Li et al. [22].

The RETOUCH dataset was made public as part of the MICCAI 2017 challenge. The goal of the RETOUCH Challenge is to compare algorithms for detecting and segmenting retinal fluids IRF, SRF, and PED from OCT scans obtained from various vendors. The data is divided into training and testing sets. Given the unavailability of testing labels, we implemented an unbiased 3-fold cross-validation method for each scanner on the training dataset [21]. Folding was performed at the subject level to ensure that B-scans from the same volume were not simultaneously present in both the training and test sets during each iteration.

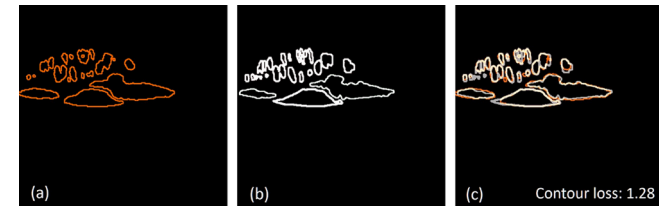


Fig. 6. The contour loss. (a) outline prediction. (b) outline label. (c) outline of the label and predict.

4.2. Evaluation metrics

The following evaluation measures are employed to compute the performance among different approaches in this work. The Dice Score (DSC), Pixel Accuracy (PA), and Balanced Accuracy (BACC) are utilized to assess the similarity between the ground truth and the pixel-wise segmentation results. The equations are as follows:

$$DSC = \frac{2|A \cap B|}{|A| + |B|} \quad (9)$$

$$PA = \frac{|A \cap B|}{|B|} \quad (10)$$

$$BACC = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (11)$$

where A represents the predicted region, and B corresponds to the ground truth. For pixel classification, TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. DSC assesses the overlap between observations, making it ideal for class imbalance, while PA calculates the ratio of correctly classified pixels, thereby offering clarity; meanwhile, BACC considers and balances category accuracy, thus enhancing fairness for imbalanced data.

4.3. Experimental details

The framework is built on PyTorch [53]. All experiments are performed with a NVIDIA RTX A4000 GPU (16 GB). In Eq. (3), the parameter ratios for $\alpha : \beta$ were configured as 1:0.2. In Eq. (4), the parameter ratios for $\lambda_1 : \lambda_2 : \lambda_3$ were configured as 0.6:0.3:0.1. All datasets use the same preprocessing strategy, yet the hyperparameter configurations differ. For the DUKE DME dataset, we resized each image to 256x256 without employing data augmentation, and a learning strategy was utilized where the learning rate decays by a factor of 0.6 every 30 epochs. For the Peripapillary OCT dataset, the dimensions of each image were resized to 496x496, and the images underwent horizontal flipping as a form of augmentation. For the RETOUCH dataset, no data augmentation was applied, and training was conducted using the 'poly' learning rate strategy. It is worth noting that the original dimensions of the Peripapillary OCT dataset, provided by Li et al. [22], are 1024x992, whereas the original dimensions of the DUKE DME dataset are 524x496. Therefore, we resize the smaller datasets (DUKE DME and RETOUCH) to 256x256 to save training resources, and for the larger dataset (Peripapillary OCT), we resize it to 496x496 to preserve more fine details of the retinal structure. Although this increases the demand for computational resources, accuracy is typically the most critical factor in medical image segmentation tasks.

5. Results and discussion

5.1. DUKE DME dataset

Table 1 presents the experimental results for the DUKE DME dataset, with comparisons drawn using DSC and BACC. The proposed method outperforms the state-of-the-art methods in the segmentation of all seven retinal layers and fluid. It achieves an overall average DSC that surpasses ReLayNet [16], Language [54], Alignment [55], U-Net [29], Y-Net [17], TCCT [56], and SA CNN [31] by 9 %, 5.3 %, 3.1 %, 5.1 %, 2.1 %, 0.3 %, and 1.6 % respectively. Table 2 presents a comparison based on the BACC. The proposed method excels in most classes, particularly in INL, ONL-ISM, ISE, and OS-RPE, with an average BACC of 0.8534, showcasing its outstanding classification capability. The TCCT performs notably well in the ILM and Fluid, with an average BACC of 0.8520, demonstrating its competitiveness. A higher BACC value indicates that the classifier can more accurately balance the classification performance of positive and negative examples when handling

Table 1

DSC on the DUKE DME dataset. The best performance is highlighted in bold, and the second-best performance is denoted with an asterisk “*”. The unit of Test time is sec/B-scan.

Method	Tissue									Test time
	ILM	NFL-IPL	INL	OPL	ONL-ISM	ISE	OS-RPE	Fluid	Mean	
Language	0.85	0.89*	0.75	0.75	0.89	0.90*	0.87*	0.39	0.786	--
Alignment	0.85	0.89*	0.75	0.74	0.90*	0.90*	0.87*	0.56	0.808	--
ReLayNet	0.84	0.85	0.70	0.71	0.87	0.88	0.84	0.30	0.749	0.24
U-Net	0.84	0.89*	0.77	0.76	0.89	0.89	0.85	0.417	0.788	0.39
Y-Net	0.86	0.89*	0.78*	0.75	0.90*	0.88	0.85	0.636	0.818	0.24
TCCT	0.88	0.91	0.79	0.79	0.90*	0.90*	0.87*	0.648	0.836*	0.33
SA CNN	0.88	0.91	0.78*	0.78*	0.90*	0.91	0.89	0.536	0.823	0.17
Proposed	0.87*	0.91	0.79	0.79	0.91	0.91	0.89	0.641*	0.839	0.26

Table 2

BAcc on the DUKE DME dataset. The best performance is highlighted in bold, and the second-best performance is denoted with an asterisk “*”.

Method	Tissue								
	ILM	NFL-IPL	INL	OPL	ONL-ISM	ISE	OS-RPE	Fluid	Mean
ReLayNet	0.8606	0.9133	0.8009	0.7788	0.9011	0.9171	0.8733	0.5476	0.8268
U-Net	0.8596	0.9314	0.8130	0.7925	0.8827	0.9252	0.8982	0.4981	0.8251
Y-Net	0.8666	0.8676	0.7333	0.7599	0.9050	0.9009	0.8372	0.4973	0.7960
TCCT	0.8990	0.8772	0.8337*	0.7953*	0.8820	0.9049	0.9006*	0.7091	0.8520
SA CNN	0.8786*	0.9048	0.8125	0.7874	0.9152*	0.9072*	0.8995	0.5188	0.8280
Proposed	0.8766	0.9056*	0.8343	0.7957	0.9224	0.9087	0.9212	0.6836*	0.8534

imbalanced datasets, thereby reducing false positive and false negative rates. This improves overall classification reliability and robustness, which is particularly important in medical image segmentation tasks for accurate diagnosis and clinical decision-making.

Fig. 7 illustrates the segmentation results obtained by various methods on DUKE DME images. As depicted in Fig. 7 (c), the proposed method exhibits the excellent visual quality, while artifacts are noticeable in other methods. We categorize these artifacts into two types:

- (1) Incorrect predictions, where a certain region is identified as another region. This is evident in U-Net (Fig. 7(e)) and ReLayNet (Fig. 7(f)), as highlighted by the white dashed circles. In these cases, two small regions containing fluid were not correctly identified, and there are erroneously identified retinal layers in the upper retinal region where they shouldn't exist.
- (2) Discontinuity of layers, where a particular retinal layer unexpectedly terminates in the horizontal direction. This issue is illustrated by the red arrows in Fig. 9 (d)-(f). Apart from the proposed method, this type of artifacts is observed in all other methods.

We attribute the improved performance in all seven retinal layers and fluid to the two modules proposed and the incorporation of anatomical prior knowledge in the framework. The segmentation results conform to the learned spatial layout, implying that retinal layers should be arranged in a vertically layered structure.

In the DUKE DME dataset, due to the small proportion of pixels occupied by the fluid, which leads to an imbalance in the data, we further compared the AUC for each class. The Fig. 8 presents the ROC curves and calculates the AUC values. As shown in the figure, the proposed method performs exceptionally well across all classes, achieving the highest AUC values in each case, demonstrating its robust segmentation capability. While other methods also perform well overall, they are inferior to the proposed method. Specifically, in the fluid, the AUC of U-Net is 0.8232, significantly lower than that of other methods. Higher AUC values indicate higher sensitivity and specificity of the classifier, with lower false positive and false negative rates. Therefore, the high AUC values of the proposed method suggest that it has significant advantages in handling imbalanced datasets and accurately segmenting retinal images, providing more reliable diagnostic support.

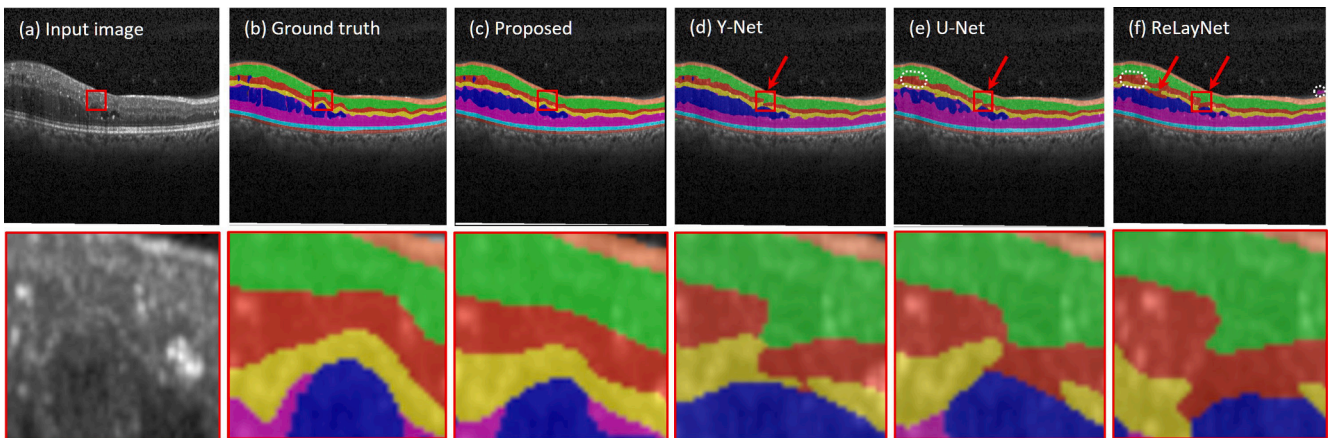


Fig. 7. Segmentation of a DUKE DME image. (a)Input image. (b)Ground truth. (c)Proposed method's prediction. (d)Y-Net's prediction. (e)U-Net's prediction. (f) ReLayNet's prediction. Incorrect predictions are marked with white dashed lines, while the discontinuity between layers is indicated with red arrows.

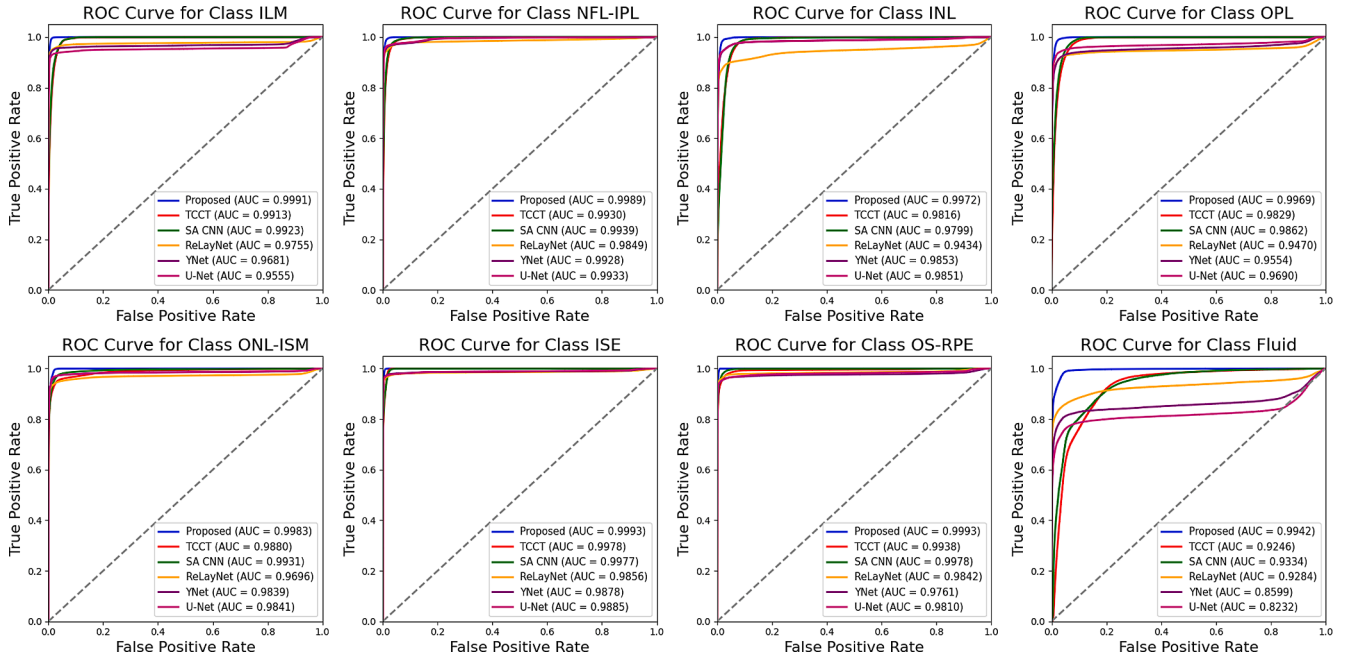


Fig. 8. The ROC curves with AUC scores for ILM, NFL-IPL, INL, OPL, ONL-ISM, ISE, OS-RPE and Fluid on the DUKE DME dataset.

5.2. Peripapillary OCT dataset

The experimental results obtained on the Peripapillary OCT dataset are presented in Table 3 and Table 4, using the same evaluation metrics as Li et al. [22], including DSC and PA. The DUKE DME dataset is centered around the macula, whereas the dataset collected by Li et al. is centered around the optic disc.

Table 3 displays the DSC for various retinal tissues. The segmentation performance of GD-Net surpasses that of the state-of-the-art methods, with a total average DSC improvement of 0.6 % across nine retinal layers and the optic disc. Specifically, in the segmentation of the optic disc, the DSC increased by 1.1 %.

In Table 4, a similar trend is observed in the PA across different tissues. Notably, GD-Net achieves an average PA exceeding that of the most advanced techniques by 0.8 % across all tissues, while the PA for the optic disc shows an enhancement of 0.8 %.

We also visualize the segmented results in Fig. 9. Compared to other segmentation methods, the proposed method exhibits superior visual quality. Consistent with our observations in Section 5.1, images segmented with U-Net and ReLayNet in Fig. 9(e) and Fig. 9(f), respectively, showed incorrect predictions on the optic disc. In other methods, artifacts such as layer discontinuities also appear, which are indicated by red arrows.

5.3. RETOUCH dataset

To validate the effectiveness of the proposed method in fluid

segmentation, a comprehensive evaluation was conducted using DSC and BAcc, comparing it with state-of-the-art fluid segmentation approaches. These methods including U-Net [29], Deeplabv3+ [57], Attention U-Net [58], CE-Net [59], RetiFluidNet [13], and also the recently developed including Curvature [14], DconnNet [60]. To ensure fairness in the evaluation process, all methods were trained using the same configuration.

The 3-fold cross-validated results on the RETOUCH challenge dataset are reported in Table 5, our model surpasses the state-of-the-art methods with average DSC and BAcc values of 87.2 % and 95.7 %, respectively. It outperforms the top-performing DconnNet by 0.4 % in DSC and 0.7 % in BAcc. Despite DconnNet's utilization of rich channel-wise directional information in the latent space for feature learning, the proposed model leverages spectral domain characteristics of OCT images to achieve superior fluid segmentation performance.

Fig. 10 compares segmentation results on RETOUCH dataset between the proposed method and three state-of-the-art networks. IRF, SRF, and PED are shown in blue, green, and red, respectively. The proposed model achieves higher visual quality, particularly in segmenting small-scale retinal fluids. Notably, RetiFluidNet exhibits PED holes (second row, fourth column), which contradicts the anatomical prior of pathological fluids. Furthermore, the other competing networks exhibit varying degrees of pixel-wise misclassifications (highlighted by white dashed lines). The proposed network uses a joint loss function considering area and contour information, enhancing resistance to non-correlated local information and speckle noise. Taken together, these results demonstrate the effectiveness and robustness of our proposed

Table 3

DSC on the peripapillary OCT dataset. The best performance is highlighted in bold, and the second-best performance is indicated with an asterisk “*”.

Method	Tissue										
	RNFL	GCL	IPL	INL	OPL	ONL	IS/OS	RPE	Choroid	Disc	Mean
U-Net [29]	0.820	0.668	0.720	0.751	0.792	0.905	0.857	0.822	0.883	0.830	0.805
ReLayNet [16]	0.816	0.686	0.730	0.773	0.808	0.904	0.862	0.824	0.877	0.759	0.804
DRUNET [51]	0.824	0.683	0.727	0.769	0.805	0.906	0.858	0.822	0.882	0.785	0.806
MGU-Net [22]	0.831	0.699*	0.749	0.789	0.820*	0.909*	0.862	0.826*	0.892*	0.821	0.820*
SA CNN [31]	0.835	0.696	0.751*	0.792*	0.822	0.910	0.863*	0.823	0.887	0.806	0.819
LightReSeg[52]	0.809	0.643	0.705	0.772	0.804	0.901	0.854	0.827	0.884	0.833*	0.803
Proposed	0.833*	0.707	0.754	0.793	0.817	0.910	0.866	0.825	0.900	0.844	0.826

Table 4

Pixel Accuracy obtained through different methods on the peripapillary OCT dataset. The best performance is highlighted in bold, and the second-best performance is indicated with an asterisk “**”.

Method	Tissue										
	RNFL	GCL	IPL	INL	OPL	ONL	IS/OS	RPE	Choroid	Disc	Mean
U-Net [29]	0.831	0.669	0.721	0.760	0.783	0.914*	0.862	0.831	0.882	0.868*	0.812
ReLayNet [16]	0.818	0.687	0.709	0.779	0.797	0.903	0.866	0.823	0.866	0.775	0.802
DRUNET [51]	0.821	0.679	0.714	0.778	0.795	0.919	0.857	0.819	0.874	0.856	0.811
MGU-Net [22]	0.846	0.703*	0.747*	0.810*	0.820	0.919	0.870*	0.830	0.892	0.861	0.830*
SA CNN [31]	0.827	0.721	0.745	0.788	0.815*	0.906	0.854	0.865*	0.912	0.795	0.823
LightReSeg[52]	0.835	0.611	0.708	0.788	0.787	0.902	0.850	0.853	0.905*	0.835	0.807
Proposed	0.836*	0.732	0.757	0.814	0.815*	0.912	0.888	0.868	0.881	0.876	0.838

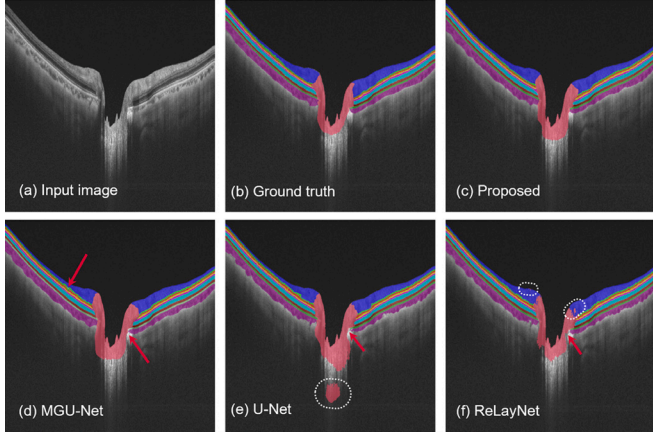


Fig. 9. Segmentation of a peripapillary OCT image. (a)Input image. (b)Ground truth. (c)Proposed method's prediction. (d)MGU-Net's prediction. (e)U-Net's prediction. (f)ReLayNet's prediction. Incorrect predictions are marked with white dashed line, while the discontinuity between layers is indicated with red arrows.

method.

Although GD-Net demonstrates excellent accuracy, we have observed that its inference speed is slower compared to some existing models such as CE-Net, DconnNet, and ReLayNet. This is primarily due

to the inclusion of complex FFE and MGC modules, which, while enhancing performance, also increase computational cost. This limitation could potentially affect the real-time processing capability in practical applications.

5.4. Experiments with joint loss function

We compared the L_{joint} , which integrates region area and contour, with several commonly used loss functions in medical image segmentation. Additionally, we evaluated it with L_{seg} , that does not incorporate L_{contour} . The results depicted in Fig. 11 demonstrate that the implementation of our introduced L_{joint} achieves varied levels of enhancement across three datasets. Specifically, on the RETOUCH dataset, the employment of L_{joint} led to a maximal increase of 5.9 % in the DSC. These outcomes suggest that the L_{joint} significantly improves the model's performance.

Fig. 12 indicates that the proposed method significantly reduces training and validation errors during the initial epochs and achieves stable performance thereafter. The minor fluctuations in the validation loss are typical and do not indicate severe overfitting. Overall, the model generalizes well to unseen data.

5.5. Selection of α , β , λ_1 , $\lambda_2\lambda_3$

In Eq. (3) and Eq. (4), the parameters α , β , λ_1 , λ_2 and λ_3 play a crucial role in balancing the contributions of different components of the loss functions. These parameters were determined through a series of

Table 5

DSC & BAcc obtained through different methods on the RETOUCH dataset. The best performance is highlighted in bold, and the second-best performance is indicated with an asterisk “**”.

Metrics	Vendor	FluidType	Method							
			U-Net	Deeplabv3+	Attention U-Net	CE-Net	RetiFulid Net	Curvature	DconnNet	Proposed
DSC	Cirrus	IRF	0.520	0.833	0.769	0.773	0.845	0.832	0.864*	0.877
		SRF	0.735	0.889	0.811	0.809	0.907	0.845	0.902*	0.894
		PED	0.727	0.845	0.743	0.749	0.854*	0.798	0.852	0.856
	Spectralis	IRF	0.614	0.744	0.684	0.666	0.824*	0.798	0.822	0.826
		SRF	0.753	0.836	0.799	0.796	0.932*	0.842	0.931	0.940
		PED	0.805	0.793	0.765	0.775	0.876	0.846	0.884	0.878*
	Topcon	IRF	0.742	0.796	0.769	0.762	0.809*	0.800	0.806	0.814
		SRF	0.851	0.847	0.854	0.815	0.879	0.864	0.882*	0.896
		PED	0.802	0.766	0.791	0.777	0.871	0.842	0.869*	0.863
	Average		0.728	0.817	0.776	0.769	0.866	0.830	0.868*	0.872
BAcc	Cirrus	IRF	0.856	0.930	0.928	0.943	0.935	0.926	0.938	0.942*
		SRF	0.904	0.964	0.959	0.959	0.967*	0.944	0.960	0.977
		PED	0.904	0.951*	0.931	0.933	0.942	0.936	0.947	0.952
	Spectralis	IRF	0.814	0.905	0.906	0.904	0.922	0.876	0.926*	0.934
		SRF	0.878	0.968	0.972	0.970	0.973	0.952	0.975*	0.977
		PED	0.907	0.933	0.940	0.943	0.956	0.954	0.959*	0.961
	Topcon	IRF	0.870	0.930	0.903	0.893	0.946*	0.884	0.934	0.954
		SRF	0.925	0.952	0.949	0.938	0.957*	0.955	0.956	0.962
		PED	0.900	0.936	0.927	0.922	0.943	0.948	0.957	0.952*
	Average		0.884	0.941	0.935	0.934	0.949	0.931	0.950*	0.957
			0.39	0.23	0.40	0.20	0.44	0.39	0.14	0.26

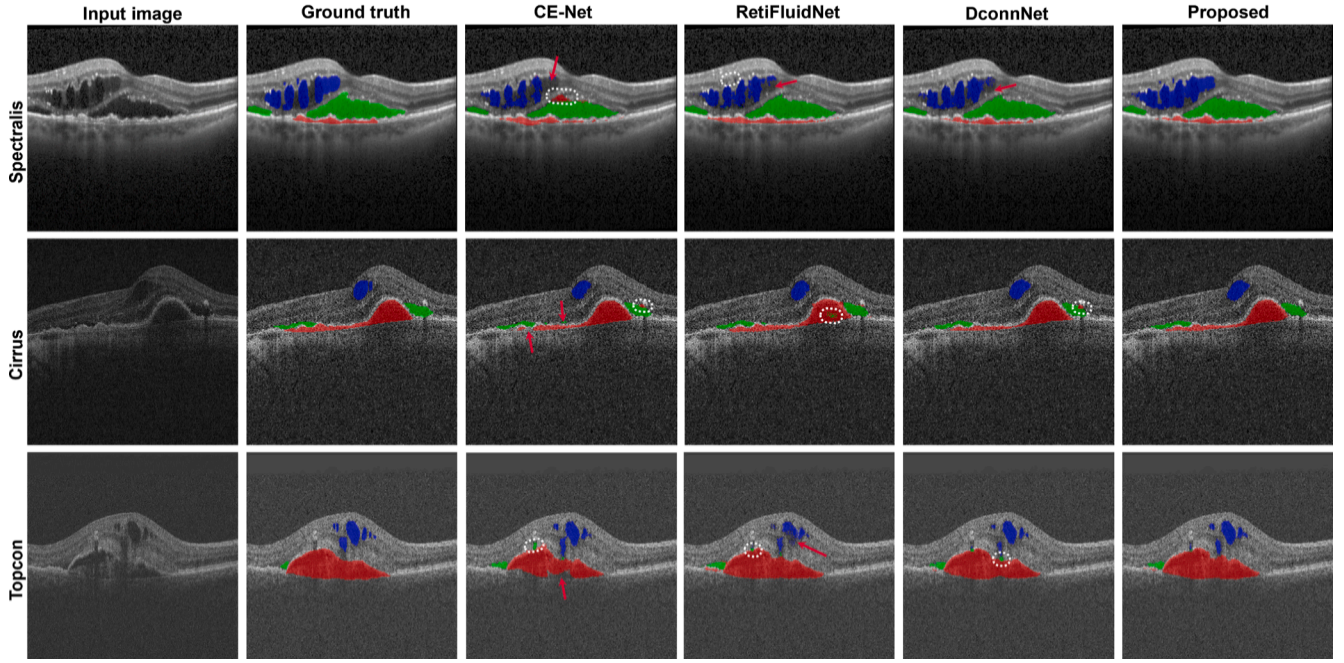


Fig. 10. Segmentation of the RETOUCH datasets. The segmentation of IRF, SRF, and PED is depicted using blue, green, and red colors, respectively. Incorrect predictions are marked with white dashed line, while discontinuities or holes within the fluid that should not exist are indicated by red arrows.

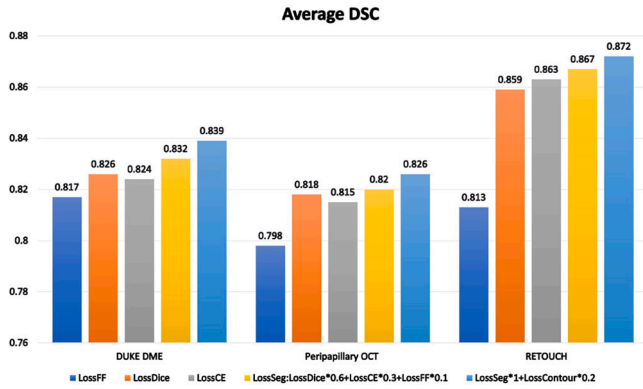


Fig. 11. Experiments with joint loss functions.

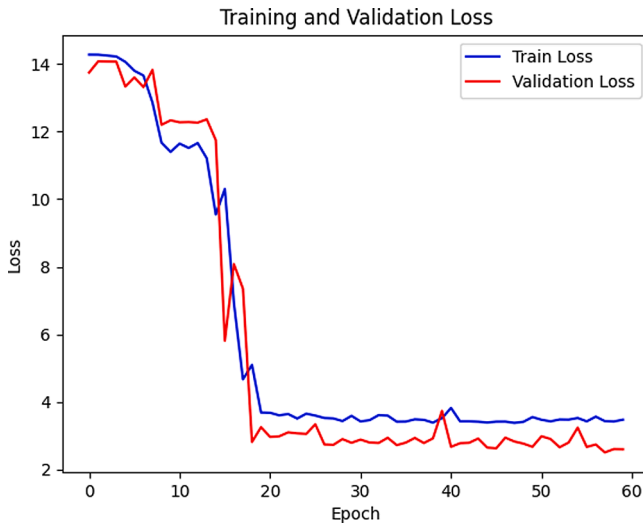


Fig. 12. Training and Validation Loss Curves on the DUKE DME Dataset.

validation experiments. First, for Eq. (4), we used the region-based L_{seg} to train the model and determined the values of λ_1 , λ_2 , and λ_3 on the DUKE DME dataset. Based on segmentation experience in medical imaging, L_{Dice} and L_{CE} have a greater impact on segmentation performance. Therefore, higher weights were assigned to them. During our experiments, we set up seven different $\lambda_1 : \lambda_2 : \lambda_3$ schemes. The specific experimental results are shown in the Table 6.

Secondly, based on the experience of adjusting parameters, we fixed the weight coefficient α of L_{seg} at 1. For the selection of β , we explored the impact of different β values on the model's performance using the DUKE DME dataset. The results are shown in the Table 7. Initially, increasing the weight of β led to better performance, with the best performance achieved at $\beta = 0.2$. When β is too small, the weight is insufficient to significantly influence the segmentation. As β increases, it positively impacts the regulation of segmentation shapes. However, when β becomes too large, the model excessively minimizes the contour loss, neglecting L_{seg} . This results in smooth but inaccurate segmentation, leading to degraded performance.

5.6. Ablation study

As illustrated in Fig. 13, to fully validate the impact of FFE and MGC on the segmentation of retinal layers and pathological fluids, we conducted a comprehensive ablation study. We evaluated the segmentation performance on retinal layers using the DUKE DME and Peripapillary OCT datasets. Given that the DUKE DME dataset also contains segmentation masks for Intraretinal Fluid (IRF), we further assessed the performance of fluid segmentation on both the DUKE DME and

Table 6

Result on DUKE DME dataset using different ratios of λ_1 , λ_2 , and λ_3 .

$\lambda_1 : \lambda_2 : \lambda_3$	DSC	PA
0.7 : 0.2 : 0.1	0.8247	0.8814
0.6 : 0.3 : 0.1	0.8324	0.8891
0.5 : 0.4 : 0.1	0.8259	0.8756
0.4 : 0.5 : 0.1	0.8257	0.8800
0.3 : 0.6 : 0.1	0.8217	0.8829
0.2 : 0.7 : 0.1	0.8209	0.8699

Table 7
Result on DUKE DME dataset with different β .

β	DSC	PA
0.01	0.8336	0.8896
0.05	0.8347	0.8796
0.1	0.8348	0.8874
0.2	0.8389	0.8976
0.4	0.8362	0.8750
0.8	0.8329	0.8714
2	0.8278	0.8642

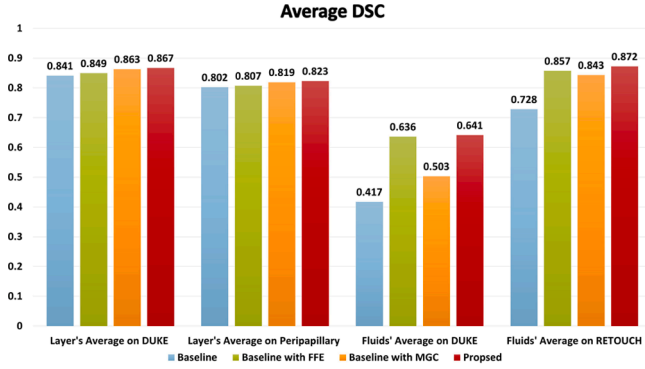


Fig. 13. Ablation experiments. The values in the figure are the corresponding average Dice score.

RETOUCH datasets. The customized U-Net model employed in our proposed framework served as the baseline to evaluate the effectiveness of FFE and MGC. The MGC is able to capture the topological structure features of the stratified arrangement between retinal layers, achieving performance enhancements of 2.2 % and 1.7 % on two datasets, respectively. Concurrently, the FFE demonstrated heightened sensitivity to the spectral components of pathological fluid in OCT images, significantly enhancing the accuracy of segmentation of fluid by 21.9 % and 12.9 % on the respective datasets. Ultimately, an examination of Fig. 11 reveals that the FFE and the MGC lead to noticeable improvements in model performance, providing empirical evidence for the effectiveness of the proposed modules.

The experiments on data preprocessing and non-preprocessing are shown in Table 8. After performing edge enhancement and histogram matching on the data, the proposed method achieved improvements of 1.02 % in DSC and 0.65 % in PA on the DUKE DME dataset. Edge enhancement increased the edge information in OCT images, enabling the model to extract edge features more efficiently. Histogram matching increased the contrast of OCT images, allowing more focus on ROI. For the multi-scale design of MGC, pooling kernels of three different sizes (2x2, 3x3, and 5x5) were used. Experiments were conducted using three different scale combination schemes, and the results are shown in Table 8. The MGC adopted the 3x3&5x5 multi-scale scheme, which achieved the best segmentation accuracy compared to the other two schemes.

The complexity analysis of the proposed model is shown in Table 9, which includes FFE and various multi-scale schemes of MGC. FFE

Table 8
Ablation Study on Data Preprocessing and Multi-Scale MGC Combination Schemes.

Data Preprocessing	MGC (3x3&5x5)	MGC (2x2&5x5)	MGC (2x2&3x3)	DSC	PA
✓	✓			0.8284	0.8469
✓	✓			0.8386	0.8534
✓		✓		0.8295	0.8473
✓			✓	0.8362	0.8401

Table 9
Complexity Analysis of FFE and MGC.

Module	FLOPs (GFLOPs)	#Params
FFE	0.21	32.69 K
MGC (3x3 & 5x5)	1.48	4.93 M
MGC (2x2 & 5x5)	2.22	5.62 M
MGC (2x2 & 3x3)	2.24	5.62 M

accounts for only a small portion of the FLOPs and parameter count. The other two scale combination schemes of MGC have higher computational complexity and parameter count, yet performance decreases.

5.7. Robustness to different OCT image vendors

To validate the robustness of the GD-Net across images acquired from different vendors, we utilized the RETOUCH dataset and conducted a comparative analysis of three types of fluid. Each box represents the average outcome derived from three different vendors. As illustrated in Fig. 14, we comprehensively compared five methods, where each method is represented by a box of a specific color, with the cross mark indicating the mean and the horizontal line representing the median. Points outside the whiskers are considered outliers. A smaller box size indicates a model's better interference resistance capability to images collected by different vendors.

Our proposed method achieved the highest DSC for the segmentation of all three types of fluids, along with the smallest box size. Notably, for the segmentation of IRF, the minimum value of the proposed method exceeded the lower quartile of other methods. No outliers were observed in the segmentation of SRF and PED.

6. Conclusion

To address the challenge of low accuracy in the simultaneous segmentation of retinal layers and fluids, we introduce a deep learning framework named GD-Net, which encompasses two modules: FFE and MGC. We trained and evaluated this model on three publicly available datasets, where the results indicate that GD-Net enhances the segmentation performance of both retinal layers and fluids compared to other state-of-the-art methods. More importantly, due to GD-Net's preservation of the retinal topological structure, it exhibits high performance in DSC, PA, and BAcc metrics, resulting in segmentation images with exceptional visual quality. This contributes to strengthening the early diagnosis of ME, alleviating vision impairment, and holds significant clinical application potential in improving ophthalmic care and enhancing patient quality of life. Future research plans involve

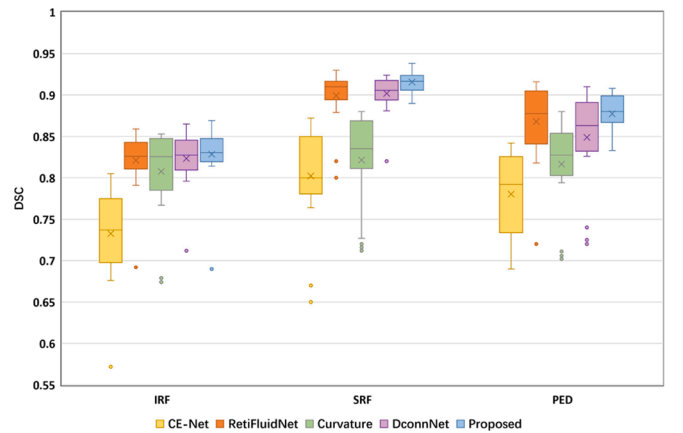


Fig. 14. Comparison of average DSC for the same type of fluid across three different vendors on the RETOUCH Dataset. Different colors represent various methods.

employing techniques such as pruning and quantization to reduce the model's parameters and runtime. These strategies aim to enhance inference speed while maintaining model performance, thereby improving its applicability in real-world scenarios. Additionally, the proposed model is intended to be applied for screening a wider range of retinal diseases, such as early-stage glaucoma. With an increase in training data, the model may be extended from two-dimensional B-scan images to three-dimensional retinal volume segmentation.

CRedit authorship contribution statement

Guogang Cao: Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition. **Zhilin Zhou:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Resources, Supervision, Validation, Writing – review & editing. **Yan Wu:** Validation, Project administration, Conceptualization. **Zeyu Peng:** Visualization, Software, Resources, Data curation. **Rugang Yan:** Software, Project administration, Data curation. **Yunqing Zhang:** Resources, Formal analysis, Data curation. **Bin Jiang:** Writing – review & editing, Validation,

Supervision, Resources, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The three datasets used are all publicly available.

Acknowledgments

The authors wish to thank the anonymous reviewers for their valuable comments and suggestions.

Supported by the National Natural Science Foundation of China (62175156), Collaborative Innovation Fund of Shanghai Institute of Technology (XTCX2022-04).

Appendix A. The dimensional changes of y_2 and y_3 in MGC

Since we use input image sizes of (256, 256) and (496, 496), the corresponding sizes in the three branches of the MGC also differ. For (256, 256), the values of y_2 and y_3 are $y_2 \in \mathbb{R}^{256 \times 5 \times 5}$ and $y_3 \in \mathbb{R}^{256 \times 3 \times 3}$ respectively. For (496, 496), the values of y_2 and y_3 are $y_2 \in \mathbb{R}^{256 \times 10 \times 10}$ and $y_3 \in \mathbb{R}^{256 \times 6 \times 6}$ respectively. It is important to note that even though the convolution kernel size is 3x3, since we set the stride and padding of all convolution operations to 1, the input and output sizes remain the same. For example, assuming the current input feature map has a shape of (B, C, H, W), the size of the feature map after convolution can be calculated using equations (A.1) and (A.2).

$$H_{out} = \frac{H + 2 \times \text{padding} - \text{kernel_size}}{\text{stride}} + 1 \quad (\text{A.1})$$

$$W_{out} = \frac{W + 2 \times \text{padding} - \text{kernel_size}}{\text{stride}} + 1 \quad (\text{A.2})$$

Substituting the parameters with padding set to 1, stride set to 1, and kernel size set to 3 into equations (A.3) and (A.4), we can obtain:

$$H_{out} = \frac{H + 2 \times 1 - 3}{1} + 1 = H \quad (\text{A.3})$$

$$W_{out} = \frac{W + 2 \times 1 - 3}{1} + 1 = W \quad (\text{A.4})$$

Therefore, after the convolution operation, the size of the feature map remains unchanged. All the scale changes occur in the max pooling and upsampling operations.

Appendix B. Experiments on contour loss alleviating the issue of internal holes in fluid

In the retinal and fluid segmentation task, internal holes within pathological fluid regions increase the loss value of the predicted lesion area. We selected the RETOUCH-Spectralis dataset for experimental validation. We recorded the contour loss value for each epoch up to 120 epochs for an image in the validation dataset. After training to convergence, we manually introduced a hole with a radius of 7 pixels within the PED region in the predicted result and observed the change in contour loss. The entire process is illustrated in the Fig. B.1:

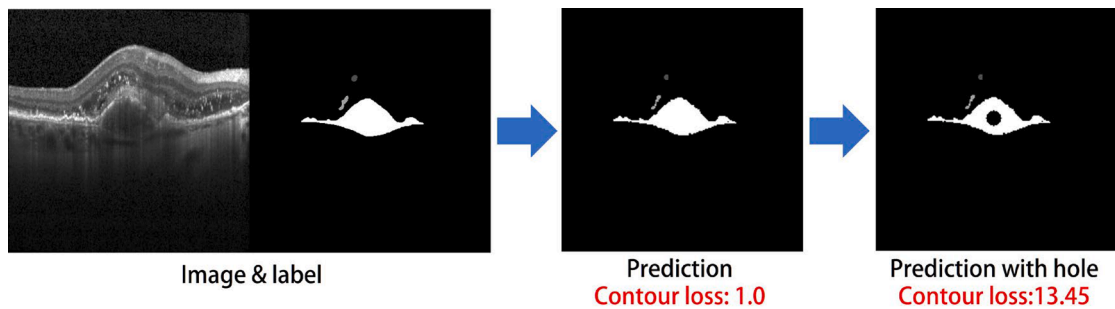


Fig. B1. Experimental Pipeline.

We observed that, upon training to convergence, the contour loss value for this feature map was 1.0. However, after introducing the hole, the

contour loss increased to 13.45. During training, we plotted only the contour loss curve. The resulting plot is shown in the Fig. B.2.

As we all know, deep learning reduces loss values through the gradient descent algorithm. Therefore, the increase in contour loss value after manually introducing a hole indicates that the contour loss is effective in mitigating the issue of internal holes in fluid regions.

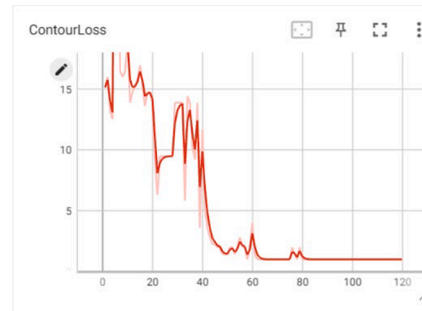


Fig. B2. Contour loss curve.

References

- [1] P.J.D.c. Romero-Aroca, "Targeting the pathophysiology of diabetic macular edema", *Am Diabetes Assoc*, 2010, pp. 2484–2485.
- [2] J.E.J.A.J.o.M.C. Kim, Exploring neovascular age-related macular degeneration and diabetic macular edema and advances in treatment, *Am. J. Manag. Care* 28 (2022).
- [3] U.N.J.A.o.M.S. Das, Diabetic macular edema, retinopathy and age-related macular degeneration as inflammatory conditions, *Arch. Med. Sci.* 12 (5) (2016) 1142–1157.
- [4] S. Saidha, et al., Primary retinal pathology in multiple sclerosis as detected by optical coherence tomography, *Brain* 134 (2) (2011) 518–533.
- [5] C.K.S. Leung, et al., Diagnostic assessment of glaucoma and non-glaucomatous optic neuropathies via optical texture analysis of the retinal nerve fibre layer, *Nat. Biomed. Eng.* 6 (5) (2022) 593–604.
- [6] J. Wu, Y. Zhang, Z. Xiao, F. Zhang, L.J.I.J.o.I.S. Geng, Technology Automated segmentation of diabetic macular edema in OCT B-scan images based on RCU-Net, *Int. J. Imaging Syst. Technol.* 33 (1) (2023) 299–311.
- [7] I.P. Marques, et al., Different risk profiles for progression of nonproliferative diabetic retinopathy: A 2-year study, *Ophthalmol Therapy* 12 (1) (2023) 485–500.
- [8] C. Gupta, et al., Choroidal structural analysis in eyes with diabetic retinopathy and diabetic macular edema—a novel OCT based imaging biomarker, *PLoS One* 13 (12) (2018) e0207435.
- [9] B. Cense, et al., Ultrahigh-resolution high-speed retinal imaging using spectral-domain optical coherence tomography, *Opt. Express* 12 (11) (2004) 2435–2447.
- [10] S. Manickam, I. R. Sheeba, and K. Venkatraman, "Study and Analysis of Fluid Filled Abnormalities in Retina Using OCT Images," 2023.
- [11] Y. He, et al., Structured layer surface segmentation for retina OCT using fully convolutional regression networks, *Med. Image Anal.* 68 (2021) 101856.
- [12] Y. He et al., "Topology guaranteed segmentation of the human retina from OCT using convolutional neural networks," *arXiv preprint arXiv:1803.05120*, 2018.
- [13] R. Rasti, A. Biglari, M. Rezapourian, Z. Yang, S.J.I.T.o.M.I. Farsi, RetiFluidNet: A self-adaptive and multi-attention deep convolutional network for retinal OCT fluid segmentation, *IEEE Trans. Med. Imaging* (2022).
- [14] G. Xing, et al., Multi-scale pathological fluid segmentation in OCT with a novel curvature loss in convolutional neural network, *IEEE Trans Med Imaging* 41 (6) (2022) 1547–1559.
- [15] Y. Tao, et al., LAGAN: Lesion-aware generative adversarial networks for edema area segmentation in SD-OCT images, *IEEE J. Biomed. Health Inform.* (2023).
- [16] A.G. Roy, et al., ReLayNet: retinal layer and fluid segmentation of macular optical coherence tomography using fully convolutional networks, *Biomed. Opt. Express* 8 (8) (2017) 3627–3642.
- [17] A. Farshad, Y. Yeganeh, P. Gehlbach, N. Navab, Y-Net: A spatio-spectral dual-encoder network for medical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 582–592.
- [18] L. Chi, B. Jiang, Y.J.A.i.N.I.P.S. Mu, "Fast fourier convolution", *Adv. Neural Inf. Proces. Syst.* 33 (2020) 4479–4488.
- [19] J. Atif, C. Hudelot, G. Fouquier, I. Bloch, E.D. Angelini, "From generic knowledge to specific reasoning for medical image interpretation using graph based representations", *IJCAI*, 2007, pp. 224–229.
- [20] S.J. Chiu, M.J. Allingham, P.S. Mettu, S.W. Cousins, J.A. Izatt, S.J.B.o.e. Farsi, Kernel regression based segmentation of optical coherence tomography images with diabetic macular edema, *Biomed. Opt. Express* 6 (4) (2015) 1172–1194.
- [21] H. Bogunović, et al., RETOUCH: The retinal OCT fluid detection and segmentation benchmark and challenge, *IEEE Trans. Med. Imaging* 38 (8) (2019) 1858–1874.
- [22] J. Li, et al., Multi-scale GCN-assisted two-stage network for joint segmentation of retinal layers and discs in peripapillary OCT images, *Biomed. Opt. Express* 12 (4) (2021) 2204–2220.
- [23] R. Kafieh, H. Rabbani, M.D. Abramoff, M.J.M.i.a. Sonka, Intra-retinal layer segmentation of 3D optical coherence tomography using coarse grained diffusion map, *Med. Image Anal.* 17 (8) (2013) 907–928.
- [24] S.J. Chiu, X.T. Li, P. Nicholas, C.A. Toth, J.A. Izatt, S.J.O.e. Farsi, Automatic segmentation of seven retinal layers in SD-OCT images congruent with expert manual segmentation, *Opt. Express* 18 (18) (2010) 19413–19428.
- [25] P.A. Dufour, et al., Graph-based multi-surface segmentation of OCT data using trained hard and soft constraints, *IEEE Trans. Med. Imaging* 32 (3) (2012) 531–543.
- [26] J. Tian, P. Marziliano, M. Baskaran, T.A. Tun, T.J.B.o.e. Aung, Automatic segmentation of the choroid in enhanced depth imaging optical coherence tomography images, *Biomed. Opt. Express* 4 (3) (2013) 397–411.
- [27] P.P. Srinivasan, S.J. Heflin, J.A. Izatt, V.Y. Arshavsky, S.J.B.o.e. Farsi, Automatic segmentation of up to ten layer boundaries in SD-OCT images of the mouse retina with and without missing layers due to pathology, *Biomed. Opt. Express* 5 (2) (2014) 348–365.
- [28] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 3431–3440.
- [29] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18, 2015, pp. 234–241: Springer.
- [30] G. Cao, et al., A single-step regression method based on transformer for retinal layer segmentation, *Phys. Med. Biol.* 67 (14) (2022) 145008.
- [31] G. Cao, Y. Wu, Z. Peng, Z. Zhou, C.J.B.O.E. Dai, Self-attention CNN for retinal layer segmentation in OCT, *Biomed. Opt. Express* 15 (3) (2024) 1605–1617.
- [32] L. Chi, G. Tian, Y. Mu, L. Xie, Q. Tian, Fast non-local neural networks with spectral residual learning, in: *Proceedings of the 27th ACM International Conference on Multimedia*, 2019, pp. 2142–2151.
- [33] G.P. Pappu, S. Tankala, K. Talabhaktula, B. Biswal, EANet: Multiscale autoencoder based edge attention network for fluid segmentation from SD-OCT images, *Int. J. Imaging Syst. Technol.* 33 (3) (2022) 909–927.
- [34] N. Rahaman, et al., On the spectral bias of neural networks, in: *International Conference on Machine Learning*, PMLR, 2019, pp. 5301–5310.
- [35] R. Suvorov, et al., Resolution-robust large mask inpainting with fourier convolutions, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2149–2159.
- [36] Y. Meng, et al., Graph-based region and boundary aggregation for biomedical image segmentation, *IEEE Trans. Med. Imaging* 41 (3) (2021) 690–701.
- [37] S.Y. Shin, S. Lee, I.D. Yun, K.M.J.M.i.a. Lee, Deep vessel segmentation by learning graphical connectivity, *Med. Image Anal.* 58 (2019) 101556.
- [38] Z. Mirikharaji and G. Hamarneh, "Star shape prior in fully convolutional networks for skin lesion segmentation," in *Medical Image Computing and Computer Assisted Intervention—MICCAI 2018: 21st International Conference, Granada, Spain, September 16–20, 2018, Proceedings, Part IV* 11, 2018, pp. 737–745: Springer.
- [39] J. J. I. T. o. p. a. Canny and m. intelligence, "A computational approach to edge detection," *IEEE Transactions on pattern analysis and machine intelligence*, no. 6, pp. 679–698, 1986.
- [40] Y. Katznelson, An introduction to harmonic analysis, Cambridge University Press, 2004.
- [41] X. Liang, Z. Hu, H. Zhang, L. Lin, E.P.J.A.i.n.i.p.s. Xing, "Symbolic graph reasoning meets convolutions", *Adv. Neural Inf. Proces. Syst.* 31 (2018).
- [42] Y. Li, A.J.A.i.n.i.p.s. Gupta, "Beyond grids: Learning graph representations for visual recognition", *Adv. Neural Inf. Proces. Syst.* 31 (2018).
- [43] Y. Chen, M. Rohrbach, Z. Yan, Y. Shuicheng, J. Feng, Y. Kalantidis, Graph-based global reasoning networks, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 433–442.
- [44] X. Li, Y. Yang, Q. Zhao, T. Shen, Z. Lin, H. Liu, Spatial pyramid based graph reasoning for semantic segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 8950–8959.

- [45] S.A. Taghanaki, et al., Combo loss: Handling input and output imbalance in multi-organ segmentation, *Comput. Med. Imaging Graph.* 75 (2019) 24–33.
- [46] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. Jorge Cardoso, “Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations,” in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: Third International Workshop, DLMIA 2017, and 7th International Workshop, ML-CDS 2017, Held in Conjunction with MICCAI 2017, Québec City, QC, Canada, September 14, Proceedings 3*, 2017, pp. 240–248: Springer.
- [47] L. Jiang, B. Dai, W. Wu, C.C. Loy, Focal frequency loss for image reconstruction and synthesis, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13919–13929.
- [48] H. Kervadec, J. Bouchtiba, C. Desrosiers, E. Granger, J. Dolz, I.B. Ayed, Boundary loss for highly unbalanced segmentation, in: *International Conference on Medical Imaging with Deep Learning*, PMLR, 2019, pp. 285–296.
- [49] A. Bokhovkin, E. Burnaev, Boundary loss for remote sensing imagery semantic segmentation, in: *International Symposium on Neural Networks*, Springer, 2019, pp. 388–401.
- [50] X. Yu, et al., Loss-balanced parallel decoding network for retinal fluid segmentation in OCT, *Comput. Biol. Med.* 165 (2023) 107319.
- [51] S.K. Devalla, et al., DRUNET: a dilated-residual U-Net deep learning network to segment optic nerve head tissues in optical coherence tomography images, *Biomed. Opt. Express* 9 (7) (2018) 3244–3265.
- [52] X. He, et al., Light-weight retinal layer segmentation with global reasoning, *IEEE Trans. Instrum. Meas.* (2024).
- [53] A. Paszke, et al., Pytorch: An imperative style, high-performance deep learning library. Presented at the *Advances in Neural Information Processing Systems*, 2019.
- [54] A. Tran, J. Weiss, S. Albarqouni, S. Faghi Roohi, and N. Navab, “Retinal layer segmentation reformulated as OCT language processing,” in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part V 23*, 2020, pp. 694–703: Springer.
- [55] H. Maier, S. Faghihroohi, N. Navab, A line to align: deep dynamic time warping for retinal OCT segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2021, pp. 709–719.
- [56] Y. Tan, et al., Retinal layer segmentation in OCT images with boundary regression and feature polarization, *IEEE Trans. Med. Imaging* (2023).
- [57] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 801–818.
- [58] O. Oktay et al., “Attention u-net: Learning where to look for the pancreas,” *arXiv preprint arXiv:1803.05120*, 2018.
- [59] Z. Gu, et al., Ce-net: Context encoder network for 2d medical image segmentation, *IEEE Trans. Med. Imaging* 38 (10) (2019) 2281–2292.
- [60] Z. Yang, S. Farsiu, Directional connectivity-based segmentation of medical images, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11525–11535.